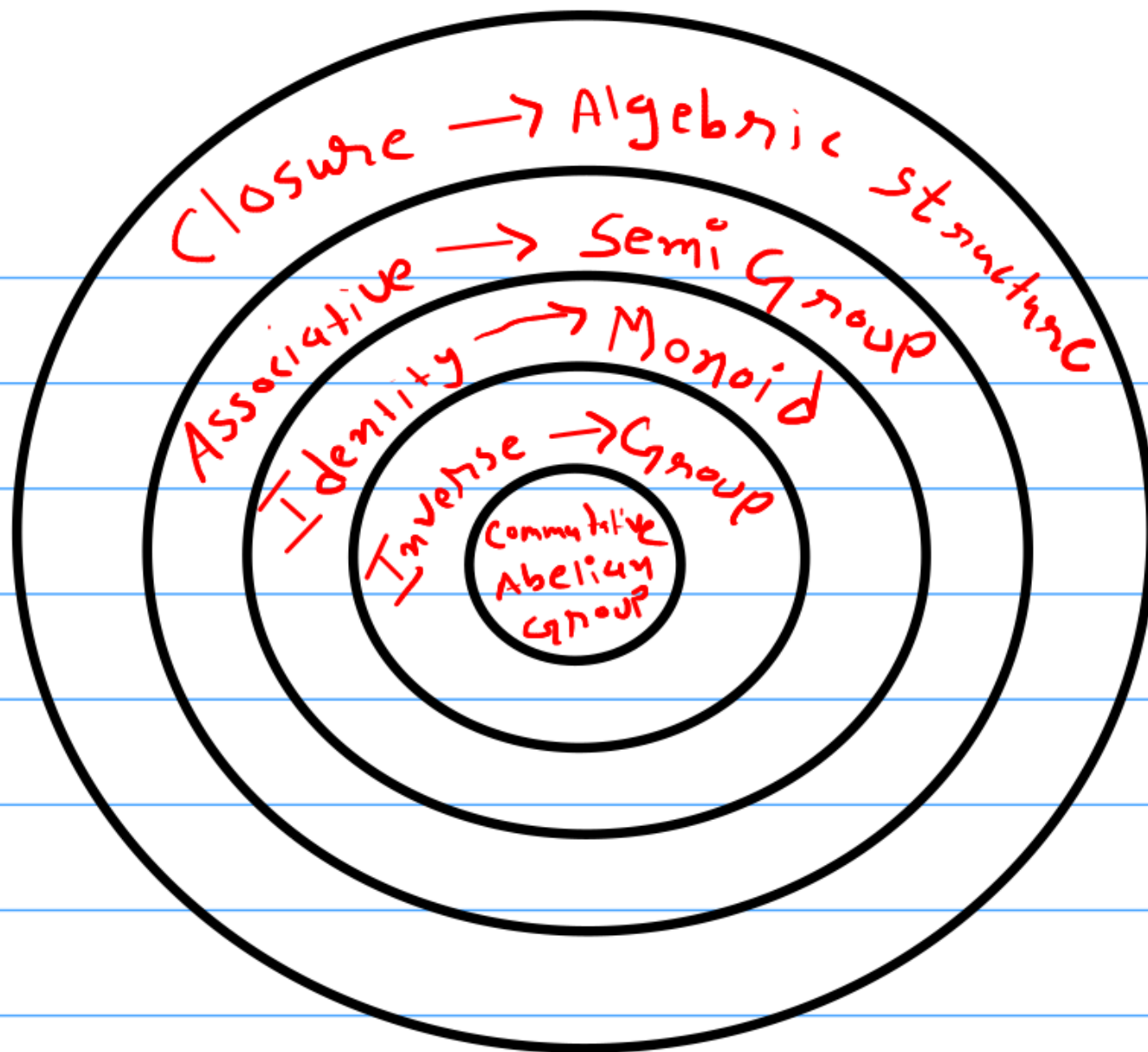


Group Theory

Introduction

- Set theory : Study of set operations.
- Relations : Study of pairs generated by product of two sets.
- Group Theory : Study of set with respect to some operation.



Algebraic Structure

- A non empty set S is called an algebraic structure w.r.t binary operation $(*)$ if it follows following axioms:
 - **Closure:** $(a*b)$ belongs to S for all $a, b \in S$.

Let $S = \{-1, 1\}$

is $(S, *)$ algebraic structure?

$$-1 \times -1 = 1$$

$$-1 \times 1 = -1$$

$$1 \times 1 = 1$$

$$1 \times -1 = -1$$

$\in S$

So, YES

Commonly used sets

- **Natural Numbers (N)**, (also called positive integers, counting numbers, or natural numbers); They are the numbers $\{1, 2, 3, 4, 5, \dots\}$
- **Integers (Z)**. This is the set of all whole numbers plus all the negatives (or opposites) of the natural numbers, i.e., $\{\dots, -2, -1, 0, 1, 2, \dots\}$
- **Rational numbers (Q)**. This is all the fractions where the top and bottom numbers are integers; e.g., $1/2, 3/4, 7/2, -4/3, 4/1$ [Note: The denominator cannot be 0, but the numerator can be].
- **Real numbers (R)**, (also called measuring numbers or measurement numbers). This includes all numbers that can be written as a decimal. This includes fractions written in decimal form e.g., 0.5, 0.75 2.35, -0.073 , 0.3333, or 2.142857. It also includes all the irrational numbers such as π , $\sqrt{2}$ etc. Every real number corresponds to a point on the number line.

Algebraic Structure Examples

	Algebraic System
$\mathbb{N}, +$	
$\mathbb{N}, -$	
$\mathbb{N}, *$	
$\mathbb{N}, /$	
$\mathbb{Z}, +$	
$\mathbb{Z}, -$	
$\mathbb{Z}, *$	
$\mathbb{Z}, /$	
$\mathbb{R}, +$	
$\mathbb{R}, -$	
$\mathbb{R}, *$	
$\mathbb{R}, /$	
$\mathbb{E}, +$	
$\mathbb{E}, *$	
$\mathbb{O}, +$	
$\mathbb{O}, *$	
$\mathbb{M}, +$	
$\mathbb{M}, *$	

	Algebraic System
$N, +$	Yes
$N, -$	No
$N, *$	Yes
$N, /$	No
$Z, +$	Yes
$Z, -$	Yes
$Z, *$	Yes
$Z, /$	No
$R, +$	Yes
$R, -$	Yes
$R, *$	Yes
$R, /$	No
$E, +$	Yes
$E, *$	Yes
$O, +$	No
$O, *$	Yes
$M, +$	Yes
$M, *$	Yes